Highlights from two astronomy problems for Lagrange's equations: the Lagrange points and the cosmological redshift

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November 3, 2026

These notes use only what is in Taylor, *Classical Mechanics*, Sections 7.1, 7.3, 7.5, 7.6 and 7.8: Hamilton's principle, Lagrange's equations, constraints and degrees of freedom, a coordinate change that depends on time, ignorable coordinates, and "no t in $\mathcal{L}$ gives a conserved Hamiltonian." The first half is the restricted three-body problem as every celestial-mechanics book sets it up (e.g. Murray and Dermott, *Solar System Dynamics*, Cambridge, 1999, Chapter 3; the Lagrangian route is the one in Goldstein, Poole and Safko, *Classical Mechanics*, 3rd ed., Section 3.12); the second half, the cosmology, follows the standard textbook route for a particle in an expanding universe (P. J. E. Peebles, *The Large-Scale Structure of the Universe*, Princeton, 1980, Chapter II). Each problem is done twice, straight through first and then at length, and the redshift is also written up as a homework problem, statement and solutions (Section 5). Both halves are Newtonian throughout, and the places where general relativity would add something are flagged.

Contents

1 The Lagrange points, straight through **2**

2 The Lagrange points: Lagrange's equations in a rotating frame **6**

3 The redshift, straight through **13**

4 The redshift: Noether in an expanding universe **16**

5 The redshift as a homework problem **19**

1 The Lagrange points, straight through

This section does the problem the way every Chapter 7 problem is done, in the order of Section 7.5's recipe (p. 258) and Examples 7.4–7.7: set the problem up, write the Lagrangian with nothing left out, state the constraints, write the Lagrangian with the constraints in, take Lagrange's equations, read off what they say. Every step is here. Section 2, which follows, is the long version, with the diagram, the numbers and the side roads; where it does a piece of algebra at length, the result is quoted here by equation number and the line or two that produce it are shown.

The idea in one paragraph. The inertial coordinates (X, Y) are the ordinary, time-independent ones; the trouble is that in them the Lagrangian carries t explicitly, because the primaries move and so $U(X, Y, t)$ changes in time, and neither of Chapter 7's tests (an ignorable coordinate, Section 7.6; no explicit t, Section 7.8) finds a conserved quantity. The move is to change to rotating coordinates (x, y), which are themselves time-dependent functions of (X, Y), the kind of generalized coordinate Section 7.3 allows and Example 7.6's spinning hoop uses. In them the primaries sit still, so $U(x, y)$ has no t, and the Hamiltonian is conserved: that is the Jacobi integral.[1] The price of time-dependent coordinates is that they are not "natural" (Section 7.3), so T picks up terms linear in the velocities and a velocity-free term, and the conserved $\mathcal{H}$ is not $T + U$ (Section 7.8). The redshift problem of Sections 3 and 4 is the mirror image: the comoving coordinate $x = r/a(t)$ is also a time-dependent change, and it makes x ignorable, but t survives in $\mathcal{L}$ through $a(t)$, so there the Hamiltonian is not conserved. One problem gains a conserved $\mathcal{H}$ from the change of coordinates; the other does not.

Step 1: set up the problem. Two bodies M_1 and M_2 (typically called "the primaries"), a distance R apart, go around their centre of mass (CM) in circles at angular speed Ω; a third body of mass m moves in their gravity. Put the origin at the CM and use inertial Cartesian coordinates (X, Y, Z) with the primaries' orbit in the plane $Z = 0$. At time t the primaries are at

$$\mathbf{r}_1(t) = r_1(-\cos\Omega t,\ -\sin\Omega t,\ 0), \qquad \mathbf{r}_2(t) = r_2(\cos\Omega t,\ \sin\Omega t,\ 0),$$
$$r_1 = \frac{M_2 R}{M_1 + M_2}, \quad r_2 = \frac{M_1 R}{M_1 + M_2}, \tag{1.1}$$

on opposite sides of the CM: "origin at the CM" means $M_1\,\mathbf{r}_1 + M_2\,\mathbf{r}_2 = 0$, i.e. $M_1 r_1 = M_2 r_2$ (the bodies are on opposite sides); putting $r_2 = R - r_1$ into it gives $(M_1 + M_2)\,r_1 = M_2 R$, and likewise for r_2. Newton's second law for M_1 fixes the rate: its acceleration is $\Omega^2 r_1$ toward the CM (uniform circular motion) and the only force on it is M_2's pull GM_1M_2/R^2, also toward the CM, so $M_1\Omega^2 r_1 = GM_1M_2/R^2$; with r_1 from (1.1) this is $M_1M_2\Omega^2 R/(M_1 + M_2) = GM_1M_2/R^2$, and cancelling M_1M_2 and multiplying by $(M_1 + M_2)/R$,

$$\Omega^2 = \frac{G(M_1 + M_2)}{R^3}, \tag{1.2}$$

Kepler's third law for the pair (this is (2.1)).

Step 2: the Lagrangian, with nothing left out. The body m has three Cartesian coordinates and, so far, no constraint, so as in Section 7.1 for a single particle, $\mathcal{L} = T - U$ with

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\big) - U(X, Y, Z, t), \qquad U = -\frac{GmM_1}{|\mathbf{r} - \mathbf{r}_1(t)|} - \frac{GmM_2}{|\mathbf{r} - \mathbf{r}_2(t)|}, \tag{1.3}$$

[1] Named for C. G. J. Jacobi; background and the usual conventions are at https://en.wikipedia.org/wiki/Jacobi_integral.

where $|\mathbf{r} - \mathbf{r}_i(t)| = \sqrt{(X - X_i(t))^2 + (Y - Y_i(t))^2 + Z^2}$ and (X_i, Y_i) are the components in (1.1). Lagrange's equations for X, Y and Z are Newton's second law with the two attractions: for X, $\partial\mathcal{L}/\partial\dot{X} = m\dot{X}$ and $\partial\mathcal{L}/\partial X = -\partial U/\partial X$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{X} = \partial\mathcal{L}/\partial X$ reads $m\ddot{X} = -\partial U/\partial X$, and likewise for Y and Z; together, $m\,\ddot{\mathbf{r}} = -\boldsymbol{\nabla}U$. Neither of Chapter 7's tests finds a conserved quantity as it stands: t appears explicitly through $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$, so Section 7.8's $\mathcal{H}$ is not constant, and no coordinate is ignorable (Section 7.6), since U depends on all three.

Step 3: the constraints. Two things are being held fixed. It pays to say which kind of "held fixed" each one is.

(i) *The primaries' motion is prescribed.* $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ are given functions of t, not coordinates to be varied: m is taken too small to disturb them (the restricted three-body problem; the footnote in Section 2). This is the same move as Example 7.6, the bead on a spinning hoop, where the hoop's angle ωt is prescribed and only the bead's θ is a coordinate. It is a constraint on the system, not on m: it removes the six coordinates of the primaries from the problem and is what put t into U.

(ii) *Motion in the orbital plane.* Set $Z = 0$. This is not a constraint in Section 7.3's sense, since no wire or surface enforces it. It is a choice of initial conditions that the equations of motion respect: U is even in Z (both distances contain Z only as Z^2), so $\partial U/\partial Z = GmM_1Z/|\mathbf{r} - \mathbf{r}_1|^3 + GmM_2Z/|\mathbf{r} - \mathbf{r}_2|^3$, which is 0 on $Z = 0$, and the Z equation $m\ddot{Z} = -\partial U/\partial Z$ gives $\ddot{Z} = 0$ there. A body that starts in the plane with $\dot{Z} = 0$ stays in it, and we look only at those motions.

So once the primaries are prescribed, m has three degrees of freedom; restricting to the planar motions of (ii) removes Z and does nothing to X and Y. Two generalized coordinates remain.

Step 4: the Lagrangian with the constraints in, in good coordinates. With $Z = 0$ the Lagrangian (1.3) loses its $\dot{Z}$ term and U loses Z:

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot{X}^2 + \dot{Y}^2\big) - U(X, Y, t). \tag{1.4}$$

The only question Chapter 7 leaves open is which two generalized coordinates to use. U depends on t only because the primaries move, so in coordinates that turn with them it will not. Section 7.3 allows generalized coordinates whose relation to the position involves t (Example 7.6 uses one), so take the rotating coordinates (2.2), $X = x\cos\Omega t - y\sin\Omega t$, $Y = x\sin\Omega t + y\cos\Omega t$. In them the primaries sit at $(x, y) = (-r_1, 0)$ and $(r_2, 0)$ for all time (put $(x, y) = (r_2, 0)$ into the rotation: $X = r_2\cos\Omega t$, $Y = r_2\sin\Omega t$, which is (1.1); $(-r_1, 0)$ likewise gives $\mathbf{r}_1(t)$), so

$$U(x, y) = -\frac{GmM_1}{\sqrt{(x + r_1)^2 + y^2}} - \frac{GmM_2}{\sqrt{(x - r_2)^2 + y^2}}, \tag{1.5}$$

with no t. For the kinetic energy, differentiate the rotation with the product rule (Ω constant) and collect the $\cos\Omega t$ and $\sin\Omega t$ terms, which is (2.4): $\dot{X} = (\dot{x} - \Omega y)\cos\Omega t - (\dot{y} + \Omega x)\sin\Omega t$ and $\dot{Y} = (\dot{x} - \Omega y)\sin\Omega t + (\dot{y} + \Omega x)\cos\Omega t$. Square and add: the cross terms $\mp 2(\dot{x} - \Omega y)(\dot{y} + \Omega x)\sin\Omega t\cos\Omega t$ cancel and $\cos^2\Omega t + \sin^2\Omega t = 1$, so $\dot{X}^2 + \dot{Y}^2 = (\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2$, and expanding the squares (as in (2.5)), $\dot{X}^2 + \dot{Y}^2 = \dot{x}^2 + \dot{y}^2 + 2\Omega(x\dot{y} - y\dot{x}) + \Omega^2(x^2 + y^2)$; putting $\frac{1}{2}m$ times this into (1.4),

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U(x, y)} \tag{1.6}$$

which is (2.3). Two remarks before differentiating. There is no explicit t in $\mathcal{L}$ any more; that was the point of the coordinate change. And the coordinates are not "natural" in Section 7.3's sense: the relation between (x, y) and (X, Y) involves t, so T is not a pure quadratic in $\dot{x}, \dot{y}$ (there are the

linear terms $m\Omega(x\dot y - y\dot x)$ and the velocity-free term $\frac12 m\Omega^2(x^2+y^2)$), exactly as for the bead on the spinning hoop.

Step 5: Lagrange's equations. From (1.6),

$$\frac{\partial\mathcal{L}}{\partial\dot x} = m(\dot x - \Omega y), \qquad \frac{\partial\mathcal{L}}{\partial x} = m\Omega\dot y + m\Omega^2 x - \frac{\partial U}{\partial x},$$
$$\frac{\partial\mathcal{L}}{\partial\dot y} = m(\dot y + \Omega x), \qquad \frac{\partial\mathcal{L}}{\partial y} = -m\Omega\dot x + m\Omega^2 y - \frac{\partial U}{\partial y}.$$

With Ω constant, $\frac{d}{dt}\,m(\dot x - \Omega y) = m\ddot x - m\Omega\dot y$ and $\frac{d}{dt}\,m(\dot y + \Omega x) = m\ddot y + m\Omega\dot x$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot q = \partial\mathcal{L}/\partial q$ for $q = x$ and $q = y$ reads

$$m\ddot x - m\Omega\dot y = m\Omega\dot y + m\Omega^2 x - \frac{\partial U}{\partial x},$$
$$m\ddot y + m\Omega\dot x = -m\Omega\dot x + m\Omega^2 y - \frac{\partial U}{\partial y}.$$

Move the $-m\Omega\dot y$ and the $+m\Omega\dot x$ from the left to the right, where each doubles its partner:

$$m\ddot x = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot y, \qquad m\ddot y = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot x, \tag{1.7}$$

which is (2.7): gravity, then a centrifugal force $m\Omega^2(x,y)$ pointing away from the axis, then a Coriolis force $2m\Omega(\dot y, -\dot x)$ perpendicular to the velocity. Chapter 9 will derive both from Newton's second law in a rotating frame; here they came out of Lagrange's equations because the coordinates rotate.

Step 6: read off results. Two, each one line of reasoning from (1.6) or (1.7).

(a) *Where can m sit still?* "Sit still in the rotating frame" means $\dot x = \dot y = 0$ and $\ddot x = \ddot y = 0$. Put those into (1.7): the Coriolis terms vanish with the velocities, the left sides vanish with the accelerations, and

$$0 = -\frac{\partial U}{\partial x} + m\Omega^2 x, \qquad 0 = -\frac{\partial U}{\partial y} + m\Omega^2 y, \qquad \text{i.e.} \qquad \boldsymbol{\nabla} U_{\text{eff}} = 0, \quad U_{\text{eff}} \equiv U - \tfrac12 m\Omega^2(x^2+y^2), \tag{1.8}$$

since $m\Omega^2 x = -\partial_x\big(-\frac12 m\Omega^2(x^2+y^2)\big)$ and likewise for y. This has five solutions: three on the x axis (L_1, L_2, L_3; the equation (2.11) and the table in Section 2) and two off it, and no others (the polar-coordinates paragraph of Section 2); Figure 1 shows all five. The two off it take three lines. With $\mathbf{r} = (x,y)$ and $\mathbf{r}_1 = (-r_1, 0)$, $\mathbf{r}_2 = (r_2, 0)$ the rotating-frame positions, at a point $\mathbf{r}$ a distance R from *both* primaries (the third corner of an equilateral triangle on M_1M_2), each primary pulls with magnitude GM_i/R^2 along the unit vector $(\mathbf{r}_i - \mathbf{r})/R$, so, dividing (1.8) by m and writing the gravity term as (magnitude)×(unit vector),

$$\begin{aligned} -\frac1m\boldsymbol{\nabla} U &= \frac{GM_1}{R^2}\,\frac{\mathbf{r}_1 - \mathbf{r}}{R} + \frac{GM_2}{R^2}\,\frac{\mathbf{r}_2 - \mathbf{r}}{R} = \frac{GM_1(\mathbf{r}_1 - \mathbf{r})}{R^3} + \frac{GM_2(\mathbf{r}_2 - \mathbf{r})}{R^3} \\ &= \frac{G}{R^3}\Big[(M_1\mathbf{r}_1 + M_2\mathbf{r}_2) - (M_1+M_2)\,\mathbf{r}\Big] = -\frac{G(M_1+M_2)}{R^3}\,\mathbf{r} = -\Omega^2\,\mathbf{r}\,, \end{aligned} \tag{1.9}$$

using $M_1\mathbf{r}_1 + M_2\mathbf{r}_2 = 0$ (origin at the CM) and then (1.2); the centrifugal term of (1.8) divided by m is $+\Omega^2\mathbf{r}$, and the two cancel. Those are L_4 and L_5, 60° ahead of and behind M_2 as seen from M_1, for any mass ratio (on M_2's orbit only in the limit $\mu \to 0$). (Jupiter's Trojans.)

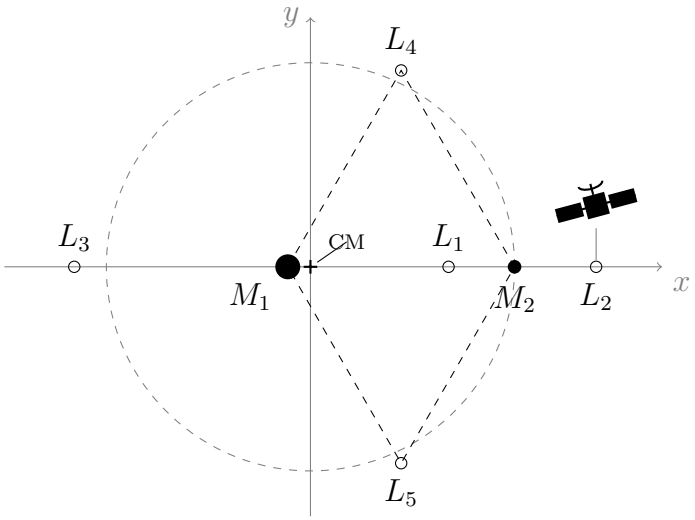


Figure 1: The five Lagrange points in the rotating frame, drawn to scale for $\mu = M_2/(M_1+M_2) = 0.1$ (the mass ratio of Figure 2; for the Sun and the Earth, L_1 and L_2 sit one percent of R from the Earth and would not separate on the page). L_1, L_2, L_3 on the line of the primaries, found from (2.11); L_4, L_5 at the third corners of the two equilateral triangles (dashed) on M_1M_2. The dashed circle is M_2's orbit about the CM. The satellite marks L_2: for the Sun and the Earth that is where JWST sits, beyond the Earth on the far side from the Sun (SOHO sits at L_1); the Trojan asteroids are at L_4 and L_5.

(b) *What is conserved?* $\mathcal{L}$ in (1.6) has no explicit t, so by Section 7.8 the Hamiltonian $\mathcal{H} = \sum_i p_i\dot{q}_i - \mathcal{L}$ is conserved. With $p_x = \partial\mathcal{L}/\partial\dot{x} = m(\dot{x} - \Omega y)$ and $p_y = \partial\mathcal{L}/\partial\dot{y} = m(\dot{y} + \Omega x)$,

$$\begin{aligned}\mathcal{H} &= m(\dot{x} - \Omega y)\dot{x} + m(\dot{y} + \Omega x)\dot{y} - \mathcal{L} = m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) - \mathcal{L} \\ &= m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) \\ &\qquad - \left[\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U\right] \\ &= \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + U_{\text{eff}}(x, y),\end{aligned} \tag{1.10}$$

the last line by subtracting (1.6) term by term: the $m\Omega(x\dot{y} - y\dot{x})$ cancel, half of $m(\dot{x}^2 + \dot{y}^2)$ survives, and $-\frac{1}{2}m\Omega^2(x^2 + y^2) + U$ is U_{eff}. This is the Jacobi integral. It is *not* $T + U$ (Section 7.8 said it would not be, because the coordinate relation involves t), and it is more useful than $T+U$ would be: since $\frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \geq 0$, a body with a given $\mathcal{H}$ can only be where $U_{\text{eff}}(x, y) \leq \mathcal{H}$. A moon with $\mathcal{H}$ below the value of U_{eff} at L_1 that starts near its planet can never leave its planet's neighbourhood, however it moves inside it.

That is the whole calculation. Section 2 adds the diagram, the numbers, the polar-coordinate route, and the question of stability, which needs the Coriolis term kept.

2 The Lagrange points: Lagrange's equations in a rotating frame

This is the long version of Section 1, with the diagram, the numbers and the side roads.

The idea in one paragraph. The inertial coordinates (X, Y) are the ordinary, time-independent ones; the trouble is that in them the Lagrangian carries t explicitly, because the primaries move and so $U(X, Y, t)$ changes in time, and neither of Chapter 7's tests (an ignorable coordinate, Section 7.6; no explicit t, Section 7.8) finds a conserved quantity. The move is to change to rotating coordinates (x, y), which are themselves time-dependent functions of (X, Y), the kind of generalized coordinate Section 7.3 allows and Example 7.6's spinning hoop uses. In them the primaries sit still, so $U(x, y)$ has no t, and the Hamiltonian is conserved: that is the Jacobi integral.[2] The price of time-dependent coordinates is that they are not "natural" (Section 7.3), so T picks up terms linear in the velocities and a velocity-free term, and the conserved $\mathcal{H}$ is not $T + U$ (Section 7.8). The redshift problem of Sections 3 and 4 is the mirror image: the comoving coordinate $x = r/a(t)$ is also a time-dependent change, and it makes x ignorable, but t survives in $\mathcal{L}$ through $a(t)$, so there the Hamiltonian is not conserved. One problem gains a conserved $\mathcal{H}$ from the change of coordinates; the other does not.

The setup. JWST sits about 1.5×10^6 km from Earth, on the far side from the Sun, at the point called L_2; Jupiter's Trojan asteroids cluster 60° ahead of and behind the planet, at L_4 and L_5. Lagrange found these five points in 1772 as special solutions of the three-body problem (the first Trojan, 588 Achilles, was not seen until 1906). In what sense is a point "at rest" when the Sun, the Earth and the point are all orbiting? In the frame that rotates with the two big bodies. Taylor does rotating frames with vectors in Chapter 9; here the same physics drops out of a Lagrangian in three lines, by the time-dependent coordinate change of Section 7.5.

Two bodies M_1 and M_2 move on circular orbits about their center of mass (CM), a distance R apart, at angular speed Ω. The CM sits on the line between them where $M_1 r_1 = M_2 r_2$ (the definition of the CM, $M_1 \mathbf{r}_1 + M_2 \mathbf{r}_2 = 0$ with the CM at the origin, the two bodies on opposite sides of it), and $r_1 + r_2 = R$; putting $r_2 = R - r_1$ into the first gives $(M_1 + M_2)\, r_1 = M_2 R$, so M_1 is $r_1 = M_2 R/(M_1 + M_2)$ from it and M_2 is $r_2 = M_1 R/(M_1 + M_2)$ (Chapter 3; Friday's two-body reduction starts the same way). What sets Ω: M_1 travels a circle of radius r_1 about the CM at angular speed Ω, so its speed is $v = \Omega r_1$ and its acceleration, toward the CM, is $v^2/r_1 = \Omega^2 r_1$ (uniform circular motion, Chapter 1). Newton's second law then says the net force on M_1 must be $M_1 \Omega^2 r_1$ toward the CM, and the only force on it is M_2's gravity, GM_1M_2/R^2, which does point at the CM. So $M_1\Omega^2 r_1 = GM_1M_2/R^2$; putting in $r_1 = M_2R/(M_1 + M_2)$ gives $M_1 M_2\, \Omega^2 R/(M_1 + M_2) = GM_1M_2/R^2$, and cancelling M_1M_2 and multiplying by $(M_1 + M_2)/R$,

$$\Omega^2 = \frac{G(M_1 + M_2)}{R^3} \tag{2.1}$$

(Kepler's third law; Chapter 8 does the general case). A third body of mass m, so small that it does not disturb the first two (this is the *restricted* three-body problem),[3] moves in their orbital

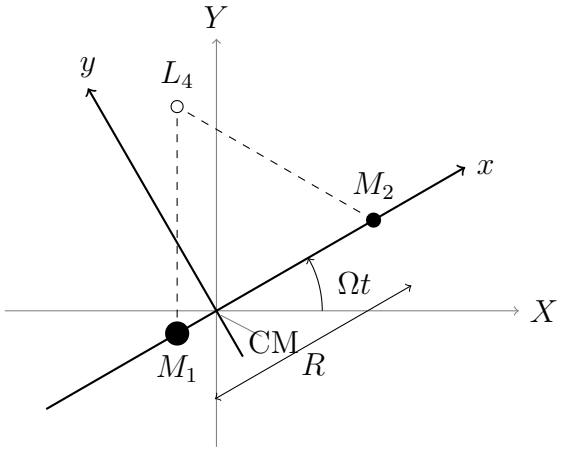
[2] Named for C. G. J. Jacobi; background and the usual conventions are at https://en.wikipedia.org/wiki/Jacobi_integral.

[3] The same approximation runs through quantum mechanics and chemistry. The hydrogen molecular ion H_2^+ with clamped nuclei is the quantum version: one light particle in the field of two heavy centres whose positions are prescribed, not solved for. There the justification is that the nuclei are slow (the Born–Oppenheimer separation of time scales); here it is that the third body's pull on the primaries is of order m/M and is dropped. Same approximation, two reasons to believe it. One more parallel: with the rotation switched off, the classical problem is

plane. Write $\mu \equiv M_2/(M_1+M_2)$, so that $r_1 = \mu R$ and $r_2 = (1-\mu)R$; the CM is at the origin with M_1 at $(-\mu R, 0)$ and M_2 at $((1-\mu)R, 0)$ in coordinates (x, y) that rotate with them:

$$X = x\cos\Omega t - y\sin\Omega t, \qquad Y = x\sin\Omega t + y\cos\Omega t, \tag{2.2}$$

where (X, Y) are inertial: the (x, y) axes are the (X, Y) axes turned counterclockwise through the angle Ωt (the standard rotation matrix), so a point at fixed (x, y) goes around the origin at angular speed Ω, in step with the primaries.



The result, first.

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2+\dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2+y^2) - U(x,y)} \tag{2.3}$$

with U the ordinary gravitational potential energy of m in the field of the two fixed primaries. Three things follow:

- Lagrange's equations are Newton's second law plus two extra forces, the centrifugal force $m\Omega^2(x, y)$ and the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$: Chapter 9's results, for free.
- A body can sit still in the rotating frame exactly where the *effective potential* $U_{\text{eff}} \equiv U - \frac{1}{2}m\Omega^2(x^2+y^2)$ has zero gradient. There are five such points (the polar-coordinates paragraph after board problem (a) shows why exactly five). Two of them, L_4 and L_5, form equilateral triangles with M_1 and M_2, for any mass ratio; the other three lie on the line through the primaries.
- $\mathcal{L}$ has no explicit t, so $\mathcal{H}$ is conserved. It is $\frac{1}{2}mv_{\text{rot}}^2 + U_{\text{eff}}$ (with $v_{\text{rot}}^2 = \dot{x}^2+\dot{y}^2$, the speed squared in the rotating frame), called the *Jacobi integral*, and it is *not* $T+U$.

Step 1: the kinetic energy. Differentiate (2.2) with the product rule (Ω is constant, so $\frac{d}{dt}\cos\Omega t = -\Omega\sin\Omega t$ and $\frac{d}{dt}\sin\Omega t = \Omega\cos\Omega t$) and collect the $\cos\Omega t$ and $\sin\Omega t$ terms. For X the product rule gives $\dot{X} = \dot{x}\cos\Omega t - \Omega x\sin\Omega t - \dot{y}\sin\Omega t - \Omega y\cos\Omega t$, and for Y, $\dot{Y} = \dot{x}\sin\Omega t + \Omega x\cos\Omega t + \dot{y}\cos\Omega t - \Omega y\sin\Omega t$; grouping,

$$\dot{X} = (\dot{x}-\Omega y)\cos\Omega t - (\dot{y}+\Omega x)\sin\Omega t, \qquad \dot{Y} = (\dot{x}-\Omega y)\sin\Omega t + (\dot{y}+\Omega x)\cos\Omega t, \tag{2.4}$$

Euler's problem of two fixed centres, which is exactly solvable, just as H_2^+ separates in prolate spheroidal coordinates. What makes the rotating problem hard is the Coriolis force, the velocity-dependent term in the equations of motion (2.7) (Step 3 of this section, Lagrange's equations).

so $(\dot X, \dot Y)$ is the pair $(\dot x - \Omega y,\ \dot y + \Omega x)$ rotated through Ωt. On squaring and adding, the cross terms $\mp 2(\dot x - \Omega y)(\dot y + \Omega x)\sin\Omega t\cos\Omega t$ cancel and $\cos^2\Omega t + \sin^2\Omega t = 1$, so the sines and cosines drop out of the speed:

$$\begin{aligned}\dot X^2 + \dot Y^2 &= (\dot x - \Omega y)^2 + (\dot y + \Omega x)^2 \\ &= (\dot x^2 - 2\Omega y\dot x + \Omega^2 y^2) + (\dot y^2 + 2\Omega x\dot y + \Omega^2 x^2) \\ &= \dot x^2 + \dot y^2 + 2\Omega\,(x\dot y - y\dot x) + \Omega^2(x^2 + y^2). \end{aligned} \tag{2.5}$$

Half of m times this is the kinetic energy in (2.3). This is Taylor's (Example 7.6) bead on a spinning hoop with two coordinates instead of one: the inertial position depends on the generalized coordinates *and* on t, and that is what breeds the extra terms.

Step 2: the potential energy. In the rotating frame the primaries do not move, so

$$U(x,y) = -\frac{GmM_1}{\sqrt{(x+\mu R)^2 + y^2}} - \frac{GmM_2}{\sqrt{(x-(1-\mu)R)^2 + y^2}} \tag{2.6}$$

has no t in it. That is the whole reason to rotate: in the inertial frame $U(X, Y, t)$ is time-dependent, and neither the energy nor the angular momentum is conserved by itself.

Step 3: Lagrange's equations. From (2.3), $\partial\mathcal{L}/\partial\dot x = m(\dot x - \Omega y)$ and $\partial\mathcal{L}/\partial x = m\Omega\dot y + m\Omega^2 x - \partial U/\partial x$; for y, $\partial\mathcal{L}/\partial\dot y = m(\dot y + \Omega x)$ and $\partial\mathcal{L}/\partial y = -m\Omega\dot x + m\Omega^2 y - \partial U/\partial y$. With Ω constant, $\frac{d}{dt}\,m(\dot x - \Omega y) = m\ddot x - m\Omega\dot y$ and $\frac{d}{dt}\,m(\dot y + \Omega x) = m\ddot y + m\Omega\dot x$, so Lagrange's equations $\frac{d}{dt}\partial\mathcal{L}/\partial\dot x = \partial\mathcal{L}/\partial x$ and $\frac{d}{dt}\partial\mathcal{L}/\partial\dot y = \partial\mathcal{L}/\partial y$ read

$$\begin{aligned} m\ddot x - m\Omega\dot y &= m\Omega\dot y + m\Omega^2 x - \partial U/\partial x, \\ m\ddot y + m\Omega\dot x &= -m\Omega\dot x + m\Omega^2 y - \partial U/\partial y. \end{aligned}$$

Moving the $-m\Omega\dot y$ and the $+m\Omega\dot x$ on the left over to the right, where each doubles its partner,

$$m\ddot x = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot y, \qquad m\ddot y = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot x. \tag{2.7}$$

Gravity, then the centrifugal force $m\Omega^2(x, y)$, outward from the axis, then the Coriolis force $2m\Omega(\dot y, -\dot x)$, perpendicular to the velocity. (Chapter 9 writes them as $-m\,\omega\times(\omega\times\mathbf{r})$ and $-2m\,\omega\times\dot{\mathbf{r}}$ with $\omega = \Omega\hat{\mathbf{z}}$; check that those give the same components.) The Coriolis force depends on velocity, so it has no potential energy, but it does no work: it is always perpendicular to $\dot{\mathbf{r}}$.

Step 4: equilibrium. To sit still in the rotating frame, $\dot x = \dot y = 0$ and $\ddot x = \ddot y = 0$. Put those into (2.7): the Coriolis terms $2m\Omega\dot y$ and $-2m\Omega\dot x$ vanish, the left sides $m\ddot x$ and $m\ddot y$ vanish, and what is left is

$$0 = -\frac{\partial U}{\partial x} + m\Omega^2 x, \qquad 0 = -\frac{\partial U}{\partial y} + m\Omega^2 y. \tag{2.8}$$

The centrifugal force came from a potential: $m\Omega^2 x = -\frac{\partial}{\partial x}\big(-\frac{1}{2}m\Omega^2(x^2+y^2)\big)$ and likewise for y. So each line is minus the derivative of $U - \frac{1}{2}m\Omega^2(x^2+y^2)$, and the two lines together say

$$\boldsymbol{\nabla} U_{\text{eff}} = 0, \qquad U_{\text{eff}}(x,y) \equiv U(x,y) - \tfrac{1}{2}m\Omega^2(x^2+y^2). \tag{2.9}$$

Notice the order: the velocities were set to zero in the *equations of motion*, after the variation, never in $\mathcal{L}$ itself. Here, as it happens, setting $\dot x = \dot y = 0$ in (2.3) first leaves $\mathcal{L} = -U_{\text{eff}}$ and the same condition, but that is luck, and the equations of motion are gone (no $\ddot x$, no Coriolis term, so

no way to ask whether the point is stable). In other problems the shortcut gives wrong equations outright: substitute the conserved angular momentum into a central-force Lagrangian before varying and the centrifugal term comes out with the wrong sign. A statement about the motion goes in after Lagrange's equations are derived; only a constraint on the *configuration* (a bead on a wire, Section 7.5) goes into the coordinates before. Figure 2 shows U_{eff} for $\mu = 0.1$ (an unrealistically heavy M_2, chosen so the five points separate on the page) with its five stationary points.

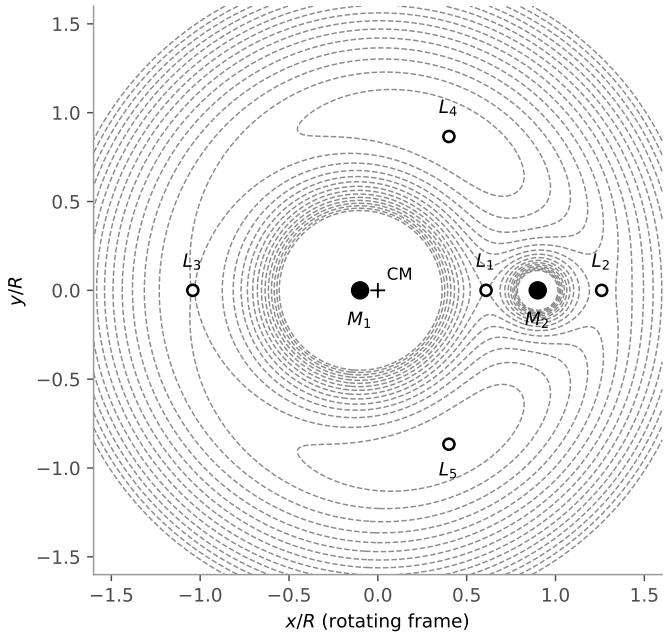


Figure 2: Contours of the effective potential U_{eff} in the rotating frame for $\mu = 0.1$, with the two primaries (filled) and the five Lagrange points (open). L_1, L_2, L_3 are saddle points; L_4 and L_5 are the two maxima. Drawn by `make_figure.py` in the handout's folder.

Board problem (a): L_4 and L_5. Show that the two points forming equilateral triangles with M_1 and M_2 satisfy (2.9), whatever the mass ratio. Work with the forces per unit mass, and use (2.1) and the fact that the CM is at the origin.

Solution. Go to class. □

Another route, for a group that finishes early: polar coordinates. Put the test mass at (r, θ) from the CM, with θ measured from the CM-to-M_2 line, so the distances to the primaries are $r_1' = \sqrt{r^2 + 2rr_1\cos\theta + r_1^2}$ and $r_2' = \sqrt{r^2 - 2rr_2\cos\theta + r_2^2}$ (law of cosines; M_1 is at angle π, M_2 at angle 0) with $r_1 = \mu R$, $r_2 = (1-\mu)R$. The centrifugal term depends on r alone, so the θ half of (2.9) is $\partial U/\partial\theta = 0$. Since $\partial r_1'/\partial\theta = -rr_1\sin\theta/r_1'$ and $\partial r_2'/\partial\theta = +rr_2\sin\theta/r_2'$, and since $U = -GmM_1/r_1' - GmM_2/r_2'$ gives, by the chain rule, $\partial U/\partial\theta = (GmM_1/r_1'^2)\,\partial r_1'/\partial\theta + (GmM_2/r_2'^2)\,\partial r_2'/\partial\theta$, we get $\partial U/\partial\theta = -GmM_1rr_1\sin\theta/r_1'^3 + GmM_2rr_2\sin\theta/r_2'^3$, and because $M_1r_1 = M_2r_2$ (the CM condition from the setup) it factors:

$$\frac{\partial U}{\partial\theta} = Gm\,r\,M_1r_1\sin\theta\left(\frac{1}{r_2'^3} - \frac{1}{r_1'^3}\right) = 0. \tag{2.10}$$

So every equilibrium has $\sin\theta = 0$ (on the axis: L_1, L_2, L_3, one in each of the three stretches the primaries cut the axis into, as "The exact positions" below shows) or $r_1' = r_2'$ (on the perpendicular bisector of the primaries: L_4, L_5), and nothing else. On the bisector $r_1' = r_2' \equiv s$, and the algebra of (a) goes through with R replaced by s: gravity per unit mass is $-G(M_1 + M_2)\,\mathbf{r}\,/s^3$, pointing at the CM, against the centrifugal $+\Omega^2\,\mathbf{r} = +G(M_1 + M_2)\,\mathbf{r}\,/R^3$. So the r half of (2.9) holds only at $s = R$, the equilateral point. This factorization is the quickest way to see that there are exactly five points. (It is how Edlund, Am. J. Phys. **92**, 414 (2024), organizes the whole problem.)

Board problem (b): L_1 and L_2 for a small μ. Use units with $R = 1$ and $G(M_1 + M_2) = 1$, so $\Omega = 1$ by (2.1), $M_1 \to 1 - \mu$ at $x = -\mu$ and $M_2 \to \mu$ at $x = 1 - \mu$. On the axis $y = 0$ the y half of (2.9) holds automatically ($\partial U/\partial y$ and $m\Omega^2 y$ both carry a factor of y), and the x half is

$$x = \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3}. \tag{2.11}$$

(This is $-\partial U_{\text{eff}}/\partial x = 0$ per unit mass at $y = 0$, with the centrifugal term $\Omega^2 x = x$ on the left and the two gravitational pulls, $-GM_i(x - x_i)/|x - x_i|^3$ toward M_i at x_i, moved to the right.) Show that for $\mu \ll 1$ both L_1 and L_2 sit a distance

$$d \approx R\left(\frac{\mu}{3}\right)^{1/3} \tag{2.12}$$

from M_2, one on each side, and evaluate it for the Sun and the Earth ($\mu = 3.00 \times 10^{-6}$, $R = 1.496 \times 10^8$ km). This d is called the *Hill radius*; inside it M_2's gravity wins over M_1's tidal pull plus the centrifugal force (the -2δ and the $+\delta$ in the solution below), which is why the Moon is Earth's.

Solution. Go to class. □

The exact positions. Equation (2.11) is a quintic once the absolute values are resolved, so the three collinear points are found numerically. There is exactly one in each of the intervals below: for f in the code (left side of (2.11) minus right side), $f'(x) = 1 + 2(1-\mu)/|x+\mu|^3 + 2\mu/|x-1+\mu|^3 > 0$, and f runs from $-\infty$ to $+\infty$ across each interval. Solving it with `scipy.optimize.brentq` on the three intervals (left of M_1, between, right of M_2):

	μ	L_1: inside M_2	L_2: beyond M_2	d from (2.12)	L_3, x/R
Sun–Earth	3.00×10^{-6}	1.49×10^6 km	1.50×10^6 km	1.50×10^6 km	-1.000001
Sun–Jupiter	9.54×10^{-4}	5.19×10^7 km	5.43×10^7 km	5.31×10^7 km	-1.0004
Earth–Moon	1.22×10^{-2}	5.80×10^4 km	6.45×10^4 km	6.13×10^4 km	-1.005

The approximation (2.12) is good to about a third of a percent for the Earth and to about five percent for the Moon, whose μ is not so small (the next term in the expansion is of relative size $\mp\frac{1}{3}(\mu/3)^{1/3}$). L_3 sits on the far side of M_1, essentially on M_2's orbit. The code is a few lines:

```
from scipy.optimize import brentq
def f(x, mu):
    return x - (1-mu)*(x+mu)/abs(x+mu)**3 - mu*(x-1+mu)/abs(x-1+mu)**3
mu = 3.003e-6
L3 = brentq(f, -2.0, -mu-1e-9, args=(mu,))
L1 = brentq(f, -mu+1e-9, 1-mu-1e-9, args=(mu,))
L2 = brentq(f, 1-mu+1e-9, 2.5, args=(mu,))
```

Noether again: the Jacobi integral. $\mathcal{L}$ in (2.3) has no explicit t, so Section 7.8 says $\mathcal{H} = p_x\dot{x} + p_y\dot{y} - \mathcal{L}$ is conserved. With $p_x = m(\dot{x} - \Omega y)$ and $p_y = m(\dot{y} + \Omega x)$,

$$\begin{aligned}\mathcal{H} &= m(\dot{x} - \Omega y)\,\dot{x} + m(\dot{y} + \Omega x)\,\dot{y} - \mathcal{L} = m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) - \mathcal{L} \\ &= m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) \\ &\qquad - \left[\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U\right] \\ &= \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + U_{\text{eff}}(x, y). \end{aligned} \tag{2.13}$$

(Subtracting (2.3), the $m\Omega(x\dot{y} - y\dot{x})$ terms cancel, half of $m(\dot{x}^2 + \dot{y}^2)$ is left, and $\tfrac{1}{2}m\Omega^2(x^2 + y^2) - U$ comes in with its sign flipped, which is U_{eff}.) This is the Jacobi integral (celestial mechanics usually quotes $C_J = -2\mathcal{H}/m$). Three remarks, the first two of which are the point of Section 7.8:

(a) It is conserved, and it looks like an energy: kinetic energy in the rotating frame plus the effective potential.

(b) It is *not* $T + U$. The true kinetic energy is $\tfrac{1}{2}m(\dot{X}^2 + \dot{Y}^2)$, and from Step 1 it is $\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2)$; adding U and subtracting (2.13),

$$\begin{aligned} T + U - \mathcal{H} = &\left[\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) + U\right] \\ &- \left[\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + U - \tfrac{1}{2}m\Omega^2(x^2 + y^2)\right] = m\Omega(x\dot{y} - y\dot{x}) + m\Omega^2(x^2 + y^2), \end{aligned} \tag{2.14}$$

which is not constant: in polar coordinates it is $\Omega\, mr^2(\dot{\theta} + \Omega)$, Ω times m's angular momentum about the CM in the inertial frame (whose angular velocity is $\dot{\theta} + \Omega$), and the primaries' pull is not aimed at the CM, so it exerts a torque. Taylor's condition for $\mathcal{H} = T + U$ (Section 7.8, pp. 269–272) was that the relation between the generalized coordinates and the inertial position not involve t; (2.2) involves t, so the condition fails, exactly as for the spinning hoop. (And $T + U$ is not conserved in the inertial frame either: there the primaries move and U depends on time. The symmetry lives in the rotating frame, and so does the conservation law.)[4]

(c) Since $\tfrac{1}{2}mv_{\text{rot}}^2 \geq 0$, a body with a given $\mathcal{H}$ can only be where $U_{\text{eff}} \leq \mathcal{H}$. The contours in Figure 2 are therefore fences (Hill's *zero-velocity curves*): a satellite with $\mathcal{H}$ below $U_{\text{eff}}(L_1)$ that starts near M_2 can never leave M_2's neighbourhood, however it moves. Raising $\mathcal{H}$ past $U_{\text{eff}}(L_1)$ opens a gate to M_1; past $U_{\text{eff}}(L_2)$, a gate to the outside.

Stability, in one paragraph (not derived today). L_1, L_2 and L_3 are saddle points of U_{eff} and are unstable: JWST fires its thrusters every few weeks to stay near L_2. L_4 and L_5 are *maxima* of U_{eff}, and yet they are stable whenever $27\mu(1 - \mu) < 1$ (a quoted result, Gascheau's of 1843; Murray and Dermott, Chapter 3, derive it), i.e. $\mu < 0.0385$, which covers Sun–Jupiter (the Trojans) and

[4] This is easy to get wrong in print. Edlund's paper (cited in the polar-coordinates paragraph) calls U_{eff} "devoid of physical meaning" because it seems to assume the body's angular velocity is constant, and proposes that the test mass's angular momentum is conserved "regionally" near each Lagrange point. Neither is needed. The Ω in U_{eff} is the *frame's* rotation rate, constant by construction; the body's own angular velocity varies, and (2.3) carries that in the kinetic and Coriolis terms. $\mathcal{H}$ in (2.13) is conserved exactly, everywhere, for the reason in Section 7.8: no t in $\mathcal{L}$. The puzzle comes from expecting a two-body potential $L^2/2mr^2$ with L the test mass's angular momentum, which is not conserved here and is not used.

Earth–Moon. A stable equilibrium at a potential maximum is impossible for a static potential: the small-oscillation analysis of Taylor's Example 7.7 would call it unstable. What rescues it is the Coriolis force, which does no work but keeps turning the velocity, so a body that starts to fall off the hill is swung around it instead (Greenberg and Davis, "Stability at potential maxima," Am. J. Phys. **46**, 1068 (1978)). Showing that means linearizing (2.7) *with* the velocity terms kept, a 4×4 eigenvalue problem; it is a very good project, and the same velocity-coupled linear equations appear in Chapter 9's Foucault pendulum.

3 The redshift, straight through

This section does the redshift problem in the order of Section 7.5's recipe (p. 258), the order Section 1 used for the Lagrange points; Section 4, which follows, is the long version, result first: set the problem up, write the Lagrangian with nothing left out, state the constraints, write the Lagrangian with the constraints in, take Lagrange's equation, read off what it says.

Step 1: set up the problem. Fill space with matter of uniform density $\rho(t)$, expanding so that every distance grows by the same factor $a(t)$ (the scale factor). Pick any point as the origin and a test particle of mass m at position $\mathbf{r} = (X, Y, Z)$, a distance $r = |\,\mathbf{r}\,|$ from it. Newton's shell theorem (the matter outside the sphere of radius r exerts no net force; the matter inside pulls as if all of it sat at the origin; that the outside cancels is the hand-wave flagged in Section 4, Step 1) and Newton's law of gravitation give the force on the particle, toward the origin,

$$F = -\frac{GmM(<r)}{r^2} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{4\pi G}{3}\,m\rho\,r, \qquad M(<r) = \tfrac{4\pi}{3}\rho r^3 \text{ the mass inside radius } r.$$

So, from $F = -\partial U/\partial r$ with $U = 0$ at the origin, its potential energy is

$$U(r,t) = m\Phi(r,t), \qquad \Phi = \frac{2\pi G}{3}\,\rho(t)\,r^2 \qquad \Big(\text{check: } -\frac{\partial U}{\partial r} = -\frac{4\pi G}{3}\,m\rho\,r = -\frac{GmM(<r)}{r^2}\Big). \tag{3.1}$$

The expansion itself obeys the same force law. A particle riding the expansion sits at $r = a(t)\,x$ with x fixed, so $\ddot r = \ddot a\,x$, and Newton's second law for that particle, $m\ddot r = -\frac{4\pi G}{3}m\rho\,r$ divided by m with $r = ax$, reads $\ddot a\,x = -\frac{4\pi G}{3}\rho\,ax$; cancelling x,

$$\ddot a = -\frac{4\pi G}{3}\,\rho\,a, \tag{3.2}$$

which is (4.5), the Friedmann acceleration equation. It is the one fact about the background we will use.

Step 2: the Lagrangian, with nothing left out. One particle with three Cartesian coordinates and no constraint yet, so as in Section 7.1, $\mathcal{L} = T - U$ with $r^2 = X^2 + Y^2 + Z^2$ in (3.1):

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot X^2 + \dot Y^2 + \dot Z^2\big) - \frac{2\pi G}{3}\,m\rho(t)\,\big(X^2 + Y^2 + Z^2\big). \tag{3.3}$$

Lagrange's equations are Newton's second law: for X, $\partial\mathcal{L}/\partial\dot X = m\dot X$ and $\partial\mathcal{L}/\partial X = -\frac{4\pi G}{3}m\rho\,X$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot X = \partial\mathcal{L}/\partial X$ reads $m\ddot X = -\frac{4\pi G}{3}m\rho\,X$, and likewise for Y and Z; together, $m\,\ddot{\mathbf{r}} = -\frac{4\pi G}{3}m\rho\,\mathbf{r}$. The time t appears explicitly through $\rho(t)$, so Section 7.8's guarantee of a constant $\mathcal{H}$ does not apply (Step 6(b) shows $\mathcal{H}$ does change), and no coordinate is ignorable, since U depends on all three. (The force is central, so the angular momentum about the origin is conserved; no Cartesian coordinate shows that, but Step 3 uses it.)

Step 3: the constraints. Two things are being held fixed, the same two kinds as in Section 1.

(i) *The background is prescribed.* $\rho(t)$ and $a(t)$ are given functions, not coordinates to be varied: the test particle does not change the universe it moves in (as the primaries' orbit is prescribed in the Lagrange-point problem, and the hoop's rotation in Example 7.6). This is what put t into $\mathcal{L}$.

(ii) *Motion along one line.* The particle has three coordinates; we use one. This is not a constraint in Section 7.3's sense, since nothing holds the particle on the line. It is a choice of initial conditions

that the equations of motion respect: the force $-\frac{4\pi G}{3}m\rho\,\mathbf{r}$ is along $\mathbf{r}$, so a particle whose velocity is along $\mathbf{r}$ keeps its velocity along $\mathbf{r}$ (the force is central, so it exerts no torque about the origin; the particle's angular momentum $m\,\mathbf{r}\times\dot{\mathbf{r}}$ is therefore conserved, and for such a particle it starts at zero). We look only at those motions.

So once the background is prescribed, the particle has three degrees of freedom; restricting to the radial motions of (ii) (put the line along the X axis, so $Y=Z=0$) removes two coordinates and does nothing to X. One generalized coordinate remains.

Step 4: the Lagrangian with the constraints in, in good coordinates. With $Y=Z=0$ the Lagrangian (3.3) loses its $\dot{Y}$ and $\dot{Z}$ terms and U loses Y and Z; writing r for X (the signed distance from the origin along the line),

$$\mathcal{L}=\tfrac{1}{2}m\dot{r}^2-\frac{2\pi G}{3}\,m\rho(t)\,r^2. \tag{3.4}$$

The only question left is which coordinate to use for the line. The t in (3.4) is there because the background expands, so use a coordinate that expands with it: the comoving coordinate $x=r/a(t)$, a time-dependent coordinate change of the kind Section 7.3 allows (Example 7.6 again). Then $r=ax$ and, by the product rule,

$$\dot{r}=\dot{a}\,x+a\,\dot{x}, \tag{3.5}$$

the Hubble flow $\dot{a}\,x=(\dot{a}/a)\,r$ (Hubble's law: the whole velocity of a particle riding the expansion, $\dot{x}=0$) plus the peculiar velocity $v_{\text{pec}}\equiv a\dot{x}$, the motion relative to the flow. Put $r=ax$ and (3.5) into (3.4) and expand the square:

$$\mathcal{L}=\tfrac{1}{2}m(\dot{a}x+a\dot{x})^2-\frac{2\pi G}{3}\,m\rho\,a^2x^2=\tfrac{1}{2}ma^2\dot{x}^2+ma\dot{a}\,x\dot{x}+\tfrac{1}{2}m\dot{a}^2x^2-\frac{2\pi G}{3}\,m\rho\,a^2x^2. \tag{3.6}$$

The cross term is the one to tame. By the product rule, $\frac{d}{dt}\big(\frac{1}{2}ma\dot{a}\,x^2\big)=\frac{1}{2}m\,\frac{d}{dt}(a\dot{a})\,x^2+\frac{1}{2}ma\dot{a}\,\frac{d}{dt}x^2=\frac{1}{2}m(\dot{a}^2+a\ddot{a})\,x^2+ma\dot{a}\,x\dot{x}$ (using the product rule again, $\frac{d}{dt}(a\dot{a})=\dot{a}^2+a\ddot{a}$, and the chain rule, $\frac{d}{dt}x^2=2x\dot{x}$), so, solving for the cross term,

$$ma\dot{a}\,x\dot{x}=\frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big)-\tfrac{1}{2}m\dot{a}^2x^2-\tfrac{1}{2}ma\ddot{a}\,x^2. \tag{3.7}$$

Put (3.7) into (3.6):

$$\mathcal{L}=\tfrac{1}{2}ma^2\dot{x}^2+\frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big)-\tfrac{1}{2}m\dot{a}^2x^2-\tfrac{1}{2}ma\ddot{a}\,x^2+\tfrac{1}{2}m\dot{a}^2x^2-\frac{2\pi G}{3}\,m\rho\,a^2x^2. \tag{3.8}$$

The two $\frac{1}{2}m\dot{a}^2x^2$ terms cancel. By (3.2), $\ddot{a}=-\frac{4\pi G}{3}\rho\,a$, so $-\frac{1}{2}ma\ddot{a}\,x^2=-\frac{1}{2}ma\big(-\frac{4\pi G}{3}\rho\,a\big)x^2=+\frac{2\pi G}{3}m\rho\,a^2x^2$, which cancels the potential term exactly. What is left is

$$\mathcal{L}=\tfrac{1}{2}ma^2\dot{x}^2+\frac{dF}{dt},\qquad F\equiv\tfrac{1}{2}ma\dot{a}\,x^2. \tag{3.9}$$

A total time derivative added to $\mathcal{L}$ changes nothing: it adds $\int_{t_1}^{t_2}(dF/dt)\,dt=F(t_2)-F(t_1)$ to the action, and since $F=\frac{1}{2}ma\dot{a}\,x^2$ depends only on x and t, which Hamilton's principle (Section 7.1) holds fixed at the endpoints, that number is the same for every path with the same endpoints, so the stationary path is the same (Problem 7.48 asks you to check this at the level of Lagrange's equations; Section 4, Step 3, does that check). So

$$\boxed{\mathcal{L}=\tfrac{1}{2}\,m\,a(t)^2\,\dot{x}^2} \tag{3.10}$$

which is (4.2). Two remarks before differentiating. The coordinate x is not "natural" in Section 7.3's sense: its relation to r involves t, and $\mathcal{L}$ carries t through $a(t)$. And x itself no longer appears: all of U cancelled, through (3.2), against the $a\ddot{a}$ term that the coordinate change produced.

Step 5: Lagrange's equation. From (3.10), $\partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ and $\partial\mathcal{L}/\partial x = 0$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x$ reads

$$\frac{d}{dt}\left(ma^2\dot{x}\right) = 0. \tag{3.11}$$

Written out with the product rule, $ma^2\ddot{x} + 2ma\dot{a}\,\dot{x} = 0$; dividing by ma^2, $\ddot{x} = -2(\dot{a}/a)\,\dot{x}$: in comoving coordinates, while the universe expands ($\dot{a} > 0$), a free particle decelerates at a rate set by the expansion.

Step 6: read off results. Two, each one line from (3.10) or (3.11).

(a) *What is conserved?* x is ignorable (Section 7.6), so its generalized momentum $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved; that is what (3.11) says. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = ma\,v_{\rm pec}$, and p is constant, so

$$v_{\rm pec} \propto \frac{1}{a}: \tag{3.12}$$

when the universe doubles in size, every freely coasting peculiar velocity halves. The argument assumed $T = \frac{1}{2}mv^2$, so by itself it says nothing about light. General relativity gives the same result for any freely moving particle, massive or not (the "Light" paragraph of Section 4): the momentum conjugate to x is conserved, and the momentum a local observer measures is that conserved momentum divided by one factor of a (here $p/a = mv_{\rm pec}$; the factor converts the comoving x to a proper distance). For a photon the measured momentum is h/λ, so $h/\lambda \propto 1/a$, $\lambda \propto a$, and

$$1 + z \equiv \frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = \frac{a_{\rm obs}}{a_{\rm emit}}, \tag{3.13}$$

the cosmological redshift.

(b) *What is not conserved?* $\mathcal{L}$ depends on t through $a(t)$, so Section 7.8's guarantee of a constant Hamiltonian does not apply, and the general result below shows that $\mathcal{H}$ does change. With $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}mv_{\rm pec}^2, \tag{3.14}$$

the kinetic energy in the peculiar velocity, and Section 7.8's general result $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ gives, since $\partial\mathcal{L}/\partial t = \frac{1}{2}m\,(2a\dot{a})\,\dot{x}^2 = ma\dot{a}\,\dot{x}^2$ (chain rule; $\mathcal{L}$ depends on t only through a),

$$\frac{d\mathcal{H}}{dt} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\cdot\tfrac{1}{2}ma^2\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H}, \qquad \text{so} \qquad \mathcal{H} \propto \frac{1}{a^2}, \tag{3.15}$$

since $d\ln\mathcal{H}/dt = -2\,d\ln a/dt$ integrates to $\ln\mathcal{H} = -2\ln a + \text{const}$. That is consistent with (3.12): $\mathcal{H} = \frac{1}{2}mv_{\rm pec}^2$ and $v_{\rm pec} \propto 1/a$. The energy is not conserved, and the formalism says exactly why: the expanding universe has no time-translation symmetry, so Noether's theorem has nothing to give. Where the energy "goes" is the subject of the last paragraph of Section 4.

That is the whole calculation.

4 The redshift: Noether in an expanding universe

This is the long version of Section 3, result first, with the side roads.

The setup. Hubble's law says a galaxy at distance r recedes from us at $v = (\dot{a}/a)\, r$, where $a(t)$ is the *scale factor*: every cosmological distance grows as $r(t) = a(t)\, x$, with x a label that does not change. Coordinates that expand with the universe like this are called *comoving coordinates*: a galaxy that just rides the expansion (the "Hubble flow") sits at a fixed x forever. Differentiating,

$$\dot{r} = \dot{a}\, x + a\, \dot{x}, \tag{4.1}$$

the Hubble flow $\dot{a}\, x = (\dot{a}/a)\, r$ (using $x = r/a$; Hubble's law: the whole motion when $\dot{x} = 0$) plus the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the motion relative to the flow (a galaxy falling toward a cluster, say, or a photon crossing the universe).

The result, first. For a particle moving freely through a homogeneous expanding universe, in comoving coordinates,

$$\boxed{\mathcal{L} = \tfrac{1}{2}\, m\, a(t)^2\, \dot{x}^2} \qquad \text{(up to a total time derivative, which changes nothing; see Step 3).} \tag{4.2}$$

Two things follow immediately from Taylor's Sections 7.6 (ignorable coordinates) and 7.8 (the Hamiltonian); both are cases of Noether's theorem, that each symmetry of $\mathcal{L}$ gives a conserved quantity:

- x does not appear in $\mathcal{L}$, so Lagrange's equation for x, $\frac{d}{dt}\,\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x = 0$, reads $\frac{d}{dt}(ma^2\dot{x}) = 0$: the coordinate is ignorable and $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = m\, a\, v_{\text{pec}}$ is constant, so

  $$v_{\text{pec}} \propto \frac{1}{a}. \tag{4.3}$$

  Random motions die away as the universe grows ("Hubble drag"): when the universe doubles in size, every freely coasting peculiar velocity halves.

- t *does* appear in $\mathcal{L}$, through $a(t)$, so there is no conserved Hamiltonian and no conserved energy (for any particle with $v_{\text{pec}} \neq 0$; one riding the flow has $\mathcal{H} = 0$ throughout). The expanding universe does not have time-translation symmetry, and Noether's theorem has nothing to give.

The rest of this section is where (4.2) comes from, and then what it says about light.

Step 1: Newton's law for the expansion. Fill space with matter of uniform density $\rho(t)$. Pick any point as the origin. A particle at r feels only the gravity of the sphere inside it (Newton's shell theorem: a uniform spherical shell pulls on a particle outside it as if all its mass sat at its center, and exerts no net force on a particle inside it; that the outside cancels is the one place Newtonian cosmology has to wave its hands, and general relativity makes the argument honest), so Newton's second law for the particle, $m\ddot{r} = -GmM(<r)/r^2$, divided by m, gives

$$\ddot{r} = -\frac{GM(<r)}{r^2} = -\frac{G}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{4\pi G}{3}\,\rho\, r, \qquad \text{i.e.} \qquad \Phi(r) = \frac{2\pi G}{3}\,\rho\, r^2. \tag{4.4}$$

Here $M(<r)$ is the mass inside radius r, and Φ is the gravitational potential energy per unit mass, chosen so that $\ddot{r} = -d\Phi/dr$. A particle riding the Hubble flow has $r = ax$ with x constant, so $\ddot{r} = \ddot{a}\, x$, and the force law above with $r = ax$ reads $\ddot{a}\, x = -\frac{4\pi G}{3}\rho\, ax$; cancelling x,

$$\ddot{a} = -\frac{4\pi G}{3}\,\rho\, a. \tag{4.5}$$

This is the Friedmann acceleration equation for a universe of pressureless matter. (General relativity's version has extra terms for pressure and for the cosmological constant; with those set to zero it is exactly this.)

Step 2: the Lagrangian in comoving coordinates. Work in one dimension; three is the same with vectors. A particle that is *not* riding the flow has $r = ax$ with $x(t)$ free. This is a time-dependent coordinate change, as in Section 7.5: $a(t)$ is a given function of time that obeys (4.5), not a coordinate, and x is the only generalized coordinate. Then

$$\begin{aligned}\mathcal{L} = \tfrac{1}{2}m\dot{r}^2 - m\Phi &= \tfrac{1}{2}m\,(\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2 \\ &= \tfrac{1}{2}ma^2\dot{x}^2 \;+\; ma\dot{a}\,x\dot{x} \;+\; \tfrac{1}{2}m\dot{a}^2x^2 \;-\; \frac{2\pi G}{3}\,m\rho\,a^2x^2. \end{aligned} \tag{4.6}$$

The cross term is the one to tame. Since $\frac{d}{dt}\left(\frac{1}{2}ma\dot{a}\,x^2\right) = \frac{1}{2}m(\dot{a}^2 + a\ddot{a})\,x^2 + ma\dot{a}\,x\dot{x}$ (product rule, with $\frac{d}{dt}(a\dot{a}) = \dot{a}^2 + a\ddot{a}$ and $\frac{d}{dt}x^2 = 2x\dot{x}$), solving for the cross term gives

$$ma\dot{a}\,x\dot{x} = \frac{d}{dt}\left(\tfrac{1}{2}ma\dot{a}\,x^2\right) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2. \tag{4.7}$$

Put that into (4.6):

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\left(\tfrac{1}{2}ma\dot{a}\,x^2\right) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2 + \tfrac{1}{2}m\dot{a}^2x^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2. \tag{4.8}$$

The two $\frac{1}{2}m\dot{a}^2x^2$ terms cancel, and by (4.5), $\ddot{a} = -\frac{4\pi G}{3}\rho\,a$, the $-\frac{1}{2}ma\ddot{a}\,x^2$ term is $-\frac{1}{2}ma\ddot{a}\,x^2 = +\frac{2\pi G}{3}m\rho a^2x^2$, which cancels the potential term exactly. What is left is

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{dF}{dt}, \qquad F \equiv \tfrac{1}{2}ma\dot{a}\,x^2. \tag{4.9}$$

Step 3: a total derivative in $\mathcal{L}$ changes nothing. The action is $S = \int_{t_1}^{t_2}\mathcal{L}\,dt$, and $\int_{t_1}^{t_2}(dF/dt)\,dt = F(t_2) - F(t_1)$ is fixed by the endpoints ($F = \frac{1}{2}ma\dot{a}\,x^2$ depends only on x and t, and Hamilton's principle, Section 7.1, varies paths with $x(t_1)$ and $x(t_2)$ held fixed): it is the same number for every path between them, so it cannot change which path makes S stationary. (Check: put $\mathcal{L} = dF/dt = \partial_tF + \dot{x}\,\partial_xF$ into Lagrange's equation by itself: $\partial\mathcal{L}/\partial\dot{x} = \partial_xF$, so $\frac{d}{dt}\,\partial\mathcal{L}/\partial\dot{x} = \partial_t\partial_xF + \dot{x}\,\partial_x^2F$, which is exactly $\partial\mathcal{L}/\partial x = \partial_x\partial_tF + \dot{x}\,\partial_x^2F$ (mixed partial derivatives are equal), and the equation reads $0 = 0$.) So we may drop it, and (4.2) is the Lagrangian. Had there been lumps in the universe, a galaxy cluster say, (4.2) would pick up $-m\phi(x,t)$ with ϕ the *peculiar* potential of the lumps alone; that is Peebles's starting point for structure formation. The smooth background's potential vanished here because the expansion already accounts for it: in Step 2 it cancelled against the $a\ddot{a}$ term, through (4.5).

The Hamiltonian, and what is not conserved. With (4.2) (the total derivative dropped) and $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}m(a\dot{x})^2 = \tfrac{1}{2}mv_{\rm pec}^2, \tag{4.10}$$

which is the kinetic energy in the peculiar velocity, and Taylor's general result $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8), with $\partial\mathcal{L}/\partial t = ma\dot{a}\,\dot{x}^2$ ($\mathcal{L}$ depends on t only through a), gives

$$\frac{d\mathcal{H}}{dt} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H}, \qquad \text{so} \qquad \mathcal{H} \propto \frac{1}{a^2}, \tag{4.11}$$

consistent with (4.3), since $\mathcal{H} = \frac{1}{2}mv_{\rm pec}^2$ and $v_{\rm pec} \propto 1/a$. (The middle step is $ma\dot{a}\,\dot{x}^2 = (2\dot{a}/a) \cdot \frac{1}{2}ma^2\dot{x}^2$, and $d\ln\mathcal{H}/dt = -2\,d\ln a/dt$ integrates to $\mathcal{H} \propto 1/a^2$.) The energy is not conserved, and the formalism tells us exactly why: $\partial\mathcal{L}/\partial t \neq 0$.

Light. The argument above assumed $\frac{1}{2}mv^2$, so by itself it says nothing about photons. General relativity makes the same statement for any freely moving particle, massive or not: the momentum measured by comoving observers falls as $1/a$. (The symmetry argument is the one you just saw. The spacetime of a homogeneous universe does not depend on the comoving position x, so the momentum conjugate to x is conserved, and converting it to what a local observer measures costs one factor of a, just as $p = ma^2\dot{x} = a\,(mv_{\rm pec})$ above.) For a photon $p = h/\lambda$, so

$$\lambda \propto a, \qquad 1 + z \equiv \frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = \frac{a_{\rm obs}}{a_{\rm emit}}. \tag{4.12}$$

A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$. That is the cosmological redshift, and it is the same conservation law as (4.3).

Where does the photon's energy go? Nowhere. $E = pc \propto 1/a$ simply decreases. The question presumes a conserved total energy, and Noether's theorem says where that would have to come from: time-translation symmetry. An expanding universe does not have it, so there is no global energy to balance the books with. (Locally, in any small freely-falling laboratory, energy is conserved as usual; it is the bookkeeping across cosmological distances that has no ledger.) Compare Taylor's Section 7.8, as Monday's slide put it: "if $\mathcal{L}$ does not depend explicitly on time, then energy is conserved." That statement run the other way, t in $\mathcal{L}$ so no conserved energy, is not a logical consequence of it and needs its own reason. The reason is the general result quoted under "The Hamiltonian" above, $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$, and here $\partial\mathcal{L}/\partial t \neq 0$. The whole universe is the example.

5 The redshift as a homework problem

The problem of Sections 3 and 4, posed the way the course's optional challenge problems are posed: the setup is given, and parts (a)–(f) walk the same road, from $\mathcal{L}(r, \dot{r}, t)$ to the redshift. The solutions follow the statement (this is the student copy; they are left out).

Challenge problem (optional): a free particle in an expanding universe. In Newtonian cosmology, space is filled with matter of uniform density $\rho(t)$, and every distance grows by one common factor: a particle that just rides the expansion sits at $r = a(t)\,x$ with x fixed, where $a(t)$ is the *scale factor* and x is its *comoving coordinate*. By Newton's shell theorem the matter inside radius r pulls a particle there as if all of it sat at the origin (and the matter outside exerts no net force), so a particle of mass m at distance r from the origin has potential energy

$$U(r,t) = \frac{2\pi G}{3}\, m\rho(t)\, r^2 \qquad \left[\text{check: } -\frac{\partial U}{\partial r} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{GmM(<r)}{r^2}\right],$$

and the scale factor obeys the Friedmann equation $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$ (Newton's second law for a particle riding the expansion, with $r = ax$; you may take it as given). Now consider a particle of mass m that moves along a line through the origin but does *not* necessarily ride the expansion.

(a) Write down the Lagrangian $\mathcal{L}(r, \dot{r}, t)$. Find the Lagrange equation for r and check that it is Newton's second law with the force above. Explain why neither of our two conservation tests (an ignorable coordinate, Section 7.6; no explicit t in $\mathcal{L}$, Section 7.8) hands you a conserved quantity here.

(b) Change to the comoving coordinate $x = r/a(t)$ (a generalized coordinate whose relation to r involves t, as in Example 7.6). Show that

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big).$$

Hint: expand the square in $\frac{1}{2}m\dot{r}^2$; write the cross term $ma\dot{a}\,x\dot{x}$ as a total time derivative minus what is left over; use the Friedmann equation on what is left over.

(c) Explain why the total time derivative can be dropped without changing the equation of motion (Problem 7.48, or argue directly from Hamilton's principle). From here on take $\mathcal{L} = \frac{1}{2}ma(t)^2\dot{x}^2$.

(d) Which coordinate is now ignorable? Find the conserved generalized momentum, and show that the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the particle's velocity relative to the expansion, falls as $1/a$. (When the universe doubles in size, a galaxy's random motion halves.)

(e) Find the Hamiltonian $\mathcal{H} = p\dot{x} - \mathcal{L}$. Is it $T + U$? Is it conserved? Use $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8) to find how $\mathcal{H}$ depends on a.

(f) General relativity gives the same conservation law for light: the momentum a local observer measures is the conserved momentum divided by a. With $p_{\text{local}} = h/\lambda$, deduce the cosmological redshift $1 + z \equiv \lambda_{\text{obs}}/\lambda_{\text{emit}} = a_{\text{obs}}/a_{\text{emit}}$.